Biochemistry Cumulative Exam | February 19, 2016

Photosynthesis by plant chloroplasts is responsible for producing all the O_2 in our atmosphere, all the food we eat, and nearly all the energy we use (generated from burning hydrocarbon fuels produced by geological processes from ancient organisms). Plant photosynthesis is carried out in chloroplasts by a collection of proteins, including photosystem I (PSI), photosystem II (PSII), cytochrome b_6f, plastocyanin, ferredoxin, ferredoxin-$NADP^+$ reductase, and ATP Synthase. The net reaction carried out by the electron transfer components is:

$$2\ H_2O + 2\ NADP^+ + 10\ H^+_{stroma} + 8\ \text{photons} \rightarrow O_2 + 2\ NADPH + 12\ H^+_{lumen} \qquad (1)$$

where stroma is the aqueous space in the chloroplast outside the thylakoid membrane, and lumen in the aqueous space inside the thylakoid membrane. Chloroplast ATP Synthase is similar to mitochondrial ATP Synthase except that it contains 14 c subunits in its F_o rotor. The following parameters were measured in a chloroplast undergoing active photosynthesis:

$pH_{stroma} - pH_{lumen} = 4.0$; $V_{stroma} - V_{lumen} = 0$

In stroma: $[ATP] = 10$ mM, $[P_i] = 10$ mM, $[ADP] = 1$ mM

$ADP + P_i \rightarrow ATP + H_2O$: $\Delta G^{o\prime} = +30.5$ kJ/mol

$NADP^+ + 2\ e^- + H^+ \rightarrow NADPH$: $E_o' = -0.32$ V ; $\frac{1}{2} O_2 + 2\ e^- + 2\ H^+ \rightarrow H_2O$: $E_o' = +0.82$ V

Questions:

1) (30 points) Draw a detailed diagram showing the arrangement of all the components of photosynthesis in the thylakoid membrane, the sites of photon capture, the sequence and stoichiometry of electron transfer and proton pumping, and the synthesis of ATP.

2) (30 points) Briefly describe the mechanism of ATP Synthase. Determine the number of protons transported per ATP synthesized by the ATP synthase, and the number of ATP synthesized per O_2 produced. Write a final net balanced equation for photosynthesis, including both the electron transfer reaction and the synthesis of ATP.

3) (40 points) Carry out a thermodynamic analysis of photosynthesis as follows: A) Calculate the total energy of the moles of photons needed (ΔG for 1 mole of 600 nm photons = 200 kJ). B) Calculate the $\Delta G^{o\prime}$ for the net electron transfer reaction: $2\ H_2O + 2\ NADP^+ \rightarrow O_2 + 2\ NADPH$. C) Calculate the ΔG for proton transport, assuming that 12 protons are transported from the stroma to the lumen during photosynthetic electron transfer (Equation 1 above). D) Calculate the ΔG for ATP synthesis under the above conditions. F) Is there enough energy of proton translocation to drive ATP synthesis? Calculate the efficiency of conversion of light energy to the energy of electron transport and ATP synthesis.

Possibly useful formulas:

$\Delta G^o = -RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$; $\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

Membrane Transport: $\Delta G(X_{in} \rightarrow X_{out}) = RT\ln([X_{out}]/[X_{in}]) + ZF(V_{out} - V_{in})$.

Z = charge on X

R = 8.3 Joule/degree.mol, F = 96.485 KJoule/volt.mol, T = 298 °K